

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 21 Nutrient Cycles

21.1 Multiple Choice Questions

1) _____ is a term used to describe dead organic matter.

- A) Humus
- B) Loam
- C) Pyrite
- D) Xenocompost

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

2) Which of the following is/are NOT a degradation product of organic material?

- A) carbon dioxide
- B) methane
- C) oxygen
- D) All of these answer choices are degradation products of organic material.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

3) Which organisms form the foundation of the carbon cycle?

- A) anaerobic chemolithotrophs
- B) chemoheterotrophs
- C) chemoorganotrophs
- D) phototrophs

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

4) Frozen methane molecules are called

- A) anhydrous methane.
- B) anoxic methane.
- C) methane anhydrates.
- D) methane hydrates.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

5) A process where different microbial taxa work in cooperation to degrade a compound that neither can perform entirely on their own is called

- A) allotrophy.
- B) ammensalism.
- C) commensalism.
- D) syntrophy.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2

6) The primary energy source for termites comes from

- A) acetogenic *Archaea*.
- B) acetogenic *Bacteria*.
- C) methanogenic *Archaea*.
- D) methanogenic *Bacteria*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2

7) One important difference between the C, N, and S cycles and the P, Ca, and Si cycles is that

- A) P, Ca, and Si are NOT cycled in terrestrial environments.
- B) the P, Ca, and Si cycles do NOT involve redox changes or gaseous forms that can alter Earth's atmospheric chemistry.
- C) the P, Ca, and Si cycles do NOT involve microbial activity.
- D) the P, Ca, and Si cycles are NOT affected by global warming or other human activity.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.6

8) The metabolic process of denitrification

- A) oxidizes N_2 .
- B) oxidizes NO_3^- .
- C) reduces N_2 .
- D) reduces NO_3^- .

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

9) The metabolic process of ammonification _____ ammonia.

- A) oxidizes
- B) produces
- C) reduces
- D) transforms

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

10) Nitrate (NO_3^-) reduced into ammonia (NH_3) to produce energy for the cell is called

- A) ammonification.
- B) dissimilative reduction of nitrate to ammonia (DRNA).
- C) nitrification.
- D) redox.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

11) Which compound in fertilizer do farmers use to inhibit the production of easily leached nitrogen compounds?

- A) anhydrous ammonia
- B) carbonate
- C) nitrapyrin
- D) nitrate

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

12) The burning of fossil fuels produces significant amounts of

- A) sulfate.
- B) sulfite.
- C) sulfonate.
- D) sulfur dioxide.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

13) Which of the following is a sulfide?

- A) H_2S
- B) HS^-
- C) S^{2-}
- D) H_2S , HS^- , and S^{2-} are all sulfides.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

14) What would the result be when a soil is supplemented with S^0 ?

- A) dimethyl sulfide production due to mixotrophs
- B) organic carbon production due to anaerobes
- C) soil acidification due to chemolithotrophs
- D) syntrophic utilization of sulfite

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.4

15) When dimethylsulfoniopropionate (DMSP) is aerobically degraded in a marine environment, which product is most often used to support growth?

- A) acrylate
- B) dimethyl disulfide
- C) dimethyl sulfoxide
- D) methanethiol

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

16) Which iron species is produced primarily from iron smelting and rarely is produced by microbial-mediated processes?

- A) ferric iron (Fe^{3+})
- B) ferrous iron (Fe^{2+})
- C) ferrulic iron (Fe^{1+})
- D) iron (Fe^0)

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

17) Bacterial oxidation of Fe^{2+} is commonly studied in chemolithotrophs that thrive under which specific environmental condition?

- A) cold temperature
- B) high sunlight
- C) low (acidic) pH
- D) nutrient-replete

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

18) Diatoms require silica due to their

- A) $\text{Ca}^{2+}/\text{Si}^{4+}$ symporters.
- B) DNA replication mechanism.
- C) ornate frustules.
- D) unique phototrophic physiology.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

19) What is a characteristic of a deep sea marine cold seep/vent?

- A) acidic
- B) luminous
- C) low pressure
- D) rich in methane

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

- 20) The calcium cycle is tightly coupled to the
- A) carbon cycle through calcium carbonate formation in ocean waters.
 - B) phosphorus cycle through calcium phosphate formation in sediments and rocks.
 - C) silica cycle through calcium silicate formation in coastal ocean sediments.
 - D) nitrogen cycle because nitrogenase requires calcium ions.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.6

- 21) Phosphorus reservoirs include
- A) phosphate minerals in rocks, dissolved phosphates in aquatic environments, and the nucleic acids and phospholipids of living organisms.
 - B) mineral phosphonates and the nucleic acids and phospholipids of living organisms.
 - C) elemental phosphorus minerals and phosphonates.
 - D) elemental phosphorous, polyphosphate, and phosphonates.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

- 22) The Fe and Mn cycles include
- A) reduced and oxidized forms that are very chemically reactive.
 - B) large reservoirs of solid mineral forms that are chemically inert.
 - C) incorporation of Fe and Mn into the exoskeletons of diatoms and other alga.
 - D) toxic forms that can poison aquatic ecosystems if the cycles are imbalanced.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

- 23) Mercury is commonly found in
- A) some electronics, but not in pesticides.
 - B) some compounds that are neurotoxins, but not in electronics.
 - C) some pesticides, but not in any neurotoxins.
 - D) some electronics, some pesticides, and some neurotoxic compounds.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

- 24) Which is the most toxic form of mercury?
- A) CH_3Hg^+
 - B) Hg^0
 - C) Hg^{2+}
 - D) Hg^{4+}

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

25) What reaction is catalyzed by mercuric reductase?

- A) CH_3HgCH_3 to Hg^{2+}
- B) CH_3Hg^+ to Hg^{2+}
- C) Hg^{4+} to Hg^{2+}
- D) Hg^{2+} to Hg^0

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

26) Agricultural application of nitrogen fertilizer

- A) reduces global warming by increasing the storage of carbon dioxide in plant biomass.
- B) increases global warming by increasing microbial respiration in agricultural soils.
- C) increases global warming by increasing the production of nitrous oxide (N_2O) by denitrification in agricultural soils.
- D) has little to no effect on the carbon cycle.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.8

27) Greenhouse gases

- A) such as CO_2 , CH_4 , and N_2O trap infrared radiation, heating the Earth.
- B) such as CO_2 increase plant and animal growth, resulting in more heat production on Earth.
- C) such as CO_2 and O_3 (ozone) interact with UV light and heat the Earth.
- D) such as CO_2 , CH_4 , and N_2O destroy O_3 (ozone) and allow more UV light to enter the atmosphere, thus increasing the temperature of the Earth's surface.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.8

28) The ocean has mitigated the impact of anthropogenic carbon dioxide production by

- A) absorbing 500 billion tons of atmospheric carbon dioxide.
- B) absorbing over 80% of the trapped heat energy.
- C) storing 600 billion tons of organic carbon.
- D) absorbing 500 billion tons of atmospheric carbon dioxide and over 80% of the trapped heat energy.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.8

29) What is the major reservoir of stable nitrogen?

- A) N₂
- B) NO
- C) NO₃⁻
- D) NH₄

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

30) Which statement exemplifies the biological pump concept?

- A) Autotrophs bring in light photons to synthesize ATP.
- B) *Bacteria* oxidize Fe²⁺ into Fe³⁺, and *Archaea* reduce Fe³⁺ back into Fe²⁺.
- C) Calcium ions are brought into cells while toxic physiological by-products are released.
- D) Primary producers sink to the deep ocean, which provides deep ocean organisms with nutrients when the primary producers die.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.6

31) What is the largest carbon reservoir on Earth?

- A) terrestrial biosphere
- B) rocks and sediments
- C) aquatic biosphere
- D) fossil fuels

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

32) The largest reservoir of methane is found

- A) as methane hydrates trapped under permafrost and marine sediments.
- B) in the rumen of cows and other ruminant animals.
- C) in the atmosphere in gaseous form.
- D) trapped in volcanic rocks in tropical environments around the world.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

33) Lignin is a complex polymer of methoxylated _____ compounds and is present in _____.

- A) silica / aquatic environments
- B) mineral / rocks
- C) oil / fossil fuels
- D) aromatic / vascular plant material

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2

34) Which of the following processes does denitrification contribute to?

- A) acid rain
- B) ozone depletion
- C) global warming
- D) acid rain, ozone depletion, and global warming

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

35) The two major forms of carbon that remain following microbial degradation are

- A) methane, organic matter, and fossil fuels.
- B) methane and carbon dioxide.
- C) lignin and fossil fuels.
- D) organic matter and hydrocarbons.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.8

36) In the oceans, sulfur exists primarily as

- A) hydrogen sulfide (H_2S).
- B) elemental sulfur (S^0).
- C) sulfate (SO_4^{2-}).
- D) gypsum (CaSO_4).

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

37) Sulfide is toxic to respiring organisms, because it can combine with metals present in _____ that serve a critical function in respiration.

- A) glycolysis enzymes
- B) oxidases
- C) NADH
- D) cytochromes

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

38) Lysis of marine algae and cyanobacteria releases the major osmoregulatory solute, dimethylsulfoniopropionate (DMSP), which is degraded into the most abundant organic S compound in nature, _____.

- A) dimethyl sulfide ($\text{CH}_3\text{-S-CH}_3$)
- B) hydrogen sulfide (H_2S)
- C) gypsum (CaSO_4)
- D) elemental sulfur (S^0)

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

39) Phosphonate comprises 25% of the _____ phosphorous pool in marine environments.

- A) inorganic
- B) organic
- C) insoluble
- D) biologically unavailable

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

40) Coccolithophores and foraminiferans both use _____ to form their exoskeletons.

- A) Si^{4+}
- B) Fe^{3+}
- C) Ca^{2+}
- D) HPO_4^-

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

41) Increasing atmospheric carbon dioxide is

- A) increasing sedimentation of C, Ca, and Si.
- B) increasing the effect of the biological pump.
- C) fueling increased primary production in ocean waters.
- D) acidifying the ocean and decreasing the rate of formation of calcareous shells.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.8

42) Studying the redox cycling of Fe in the environment has led to the discovery

- A) of bacteria that transfer electrons as a form of electricity.
- B) of bacteria that destroy toxic metal ions.
- C) of archaea that have Fe in their cell walls.
- D) that Fe limits primary production in terrestrial ecosystems.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

- 43) Nitrogen fixation results in the
A) loss of biologically available N from an ecosystem.
B) formation of nitrogen gas.
C) direct formation of nitrate.
D) addition of biologically available N to an ecosystem.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.3

- 44) Nitrous oxide (N_2O) is a
A) direct product of ammonification.
B) direct product of nitrogen fixation.
C) potent greenhouse gas that is a product of nitrification.
D) potent greenhouse gas that is a product of denitrification.

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.3

- 45) Nitrification and anammox both
A) create nitrogen gas, but nitrification is aerobic while anammox is anaerobic.
B) oxidize ammonia, but nitrification uses O_2 as an electron acceptor and anammox uses NO_2^- .
C) are aerobic processes, but nitrification produces NO_3^- and anammox produces N_2 gas.
D) occur in soils at neutral pH.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.3

- 46) The oxidation of _____ occurs both biotically and abiotically, but biotic oxidation is more common at _____ pH values. pH

- A) Si / low
B) ammonia / high
C) Fe^{2+} / low
D) Hg^0 / low

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

- 47) Human activities have disrupted the carbon cycle by
- A) burning carbon stored as fossil fuels or biomass, thus increasing the amount of carbon dioxide in the atmosphere.
 - B) decreasing the primary productivity of the oceans, resulting in increased atmospheric carbon dioxide.
 - C) burning methane hydrates to produce electricity.
 - D) acidifying the ocean, resulting in the release of large amounts of dissolved carbon dioxide into the atmosphere.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

21.2 True/False Questions

- 1) In the natural carbon cycle, carbon dioxide is returned to the atmosphere by the respiration of animals and chemoorganotrophic microorganisms.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

- 2) More carbon is present in dead organic material on Earth than in living organisms.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

- 3) Humic acids are difficult to degrade, and therefore they usually remain in a system for several decades.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

- 4) Climate scientists are developing methane hydrates to decrease the effects of global warming.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

- 5) In a global sense, all nutrient cycles are coupled together.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.1

- 6) Methanogens, among other compounds, use alcohols and fatty acids to synthesize methane gas in anoxic environments.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2

7) Syntrophic organisms cannot grow fermentatively without a hydrogen-consuming partner organism.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2

8) Acetogens can ferment glucose into acetate, but methanogens cannot.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2

9) When carbon is limiting in a system, sulfate reducers often predominate the system when sufficient sulfate ions are present for reduction.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

10) The disposal of garbage or sewage in the ocean is known to cause the sudden growth of sulfate reducers.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

11) Most chemolithotrophic bacteria that oxidize sulfur are aerobes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.4

12) The precipitation of soluble iron ions (Fe^{2+} and Fe^{3+}) into solid iron occurs spontaneously in the presence of organic carbon compounds.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

13) Abiotic oxidation of Fe^{2+} requires oxygen and occurs rapidly when the pH is near 7.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.5

14) Energy yields are high enough for Fe^{2+} oxidation that the bacteria that perform this reaction require only small quantities of Fe^{2+} to be available.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.5

15) Most phosphorous, such as that found in inorganic phosphate (HPO_4^-), exist in the +5 oxidation state in nature.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

16) Silicoflagellates use silicon compounds in substrate-level phosphorylation.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

17) Carbon dioxide is the only greenhouse gas produced by anthropogenic activities.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.8

18) Ocean acidification decreases the precipitation of calcium carbonate and thus will probably affect the cycling of Ca and other nutrients in the ocean.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

19) Mercury has both organic and inorganic forms that are neurotoxic.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

20) The mercury in fish is normally in the form of elemental mercury (Hg^0).

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

21) Some bacteria have genes that confer resistance to mercury.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

22) Mercury can serve as a transcriptional activator in the regulation of certain genes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

23) Microbial activity generates methylmercury, which is much less toxic than the mercuric ion.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

24) Mercury-resistant bacteria detoxify mercury by producing Hg^0 .

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

25) Calcium is cycled most rapidly in terrestrial environments.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

26) Denitrification of fertilizer can result in harmful algal blooms.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.3

27) The two main biological activities in the Ca and Si cycles are the biosynthesis of shells and exoskeletal material as well as the mineralization of these materials in sediments.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.6

21.3 Essay Questions

1) Using the sudden bloom of cyanobacteria in a lake due to anthropogenic fertilizer inputs as an example, explain the coupling of carbon, nitrogen, and oxygen cycling. Exclude discussion of all organisms other than microorganisms.

Answer: An increase in primary producers results in more CO_2 being taken (i.e., fixed) from the air. Increased biomass growth will also make more oxygen available for respiring heterotrophic microorganisms in the lake. The cyanobacteria might also increase nitrogen fixation, which would take nitrogen out of the atmosphere and bring in bioavailable nitrogen substrates for other microorganisms to proliferate.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.8

2) What are the similarities and differences between the Fe cycle and Ca cycle?

Answer: Both Fe and Ca cycle between insoluble and soluble forms. Ca, however, has only one redox state and cycles between insoluble CaCO_3 and dissolved Ca^{2+} , while Fe cycles between oxidized Fe^{3+} and reduced Fe^{2+} . Most forms of Fe^{3+} are insoluble minerals and Fe^{2+} is soluble. Consequently, the microbial activities involved in the cycling of Fe and Ca are different. Reduction and oxidation of Fe are coupled to respiratory processes that generate energy for growth, while Ca primarily plays a biosynthetic role in the biology of diatoms. The Ca is released from the diatom exoskeleton during mineralization (decomposition) of dead diatoms.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.5, 21.6

3) Coal contains substantial amounts of mercury. When coal is burned to generate heat or electricity, the mercury is released into the atmosphere. Most of this mercury is eventually deposited in the ocean or other aquatic environments. What happens to the mercury once it is deposited in the ocean?

Answer: Answers could include a diagram similar to that shown in Figure 20.16 (page 645). In aquatic environments, bacteria convert mercury (Hg^{2+}) into methylmercury (CH_3Hg^+) or dimethylmercury ($\text{CH}_3\text{Hg}^+ \text{CH}_3$), which are extremely toxic and absorb through the skin into the flesh of aquatic organisms (such as fish or sharks). These compounds bioaccumulate and are concentrated as they pass up the food chain, potentially causing problems for humans that eat large quantities of fish (such as large tuna, salmon, or sharks). Bacteria in anoxic environments can detoxify the methylmercury compounds where they could be buried as mercuric sulfide (HgS). If the sediments are oxidized, the mercury will be released back into the overlying water.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.7

4) Why is the water solubility of different nitrogen species of importance to agriculturalists? What processes in the nitrogen cycle are LEAST beneficial to agricultural production of crops?

Answer: If a particular nitrogen species is highly water soluble, it will wash away during precipitation. The crops will then not receive the fertilizer as intended, and runoff will negatively affect any surrounding bodies of water by providing a sudden increase in nitrogen, perhaps leading to harmful algal blooms. Nitrification and denitrification are the two least beneficial processes in the nitrogen cycle because they result in a loss of biologically available nitrogen. Nitrification produces nitrate, which is highly soluble and does NOT bind to soil particles very well and is thus washed rapidly out of the soil. Denitrification is the conversion of nitrate to nitrogen gas, which also removes nitrogen from the soil. Loss of nitrogen from the soil means that less is available to the crops, which could potentially limit production.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.3

5) How would Si depletion in the photic zone of the open ocean affect the biological pump and the C cycle?

Answer: Si is required for diatoms to fix carbon dioxide into organic carbon, thus Si depletion would reduce primary production. The biological pump is the sinking and sequestration of organic carbon into deep ocean water and sediments. Si depletion would slow down the biological pump and would reduce the amount of carbon dioxide removed from the atmosphere and stored in deep ocean water and sediments, resulting in increased atmospheric carbon dioxide.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.6

6) Marine wetlands (salt marshes) are highly productive coastal ecosystems that produce a lot of organic carbon, similar to freshwater wetlands; however, salt marshes smell very different than freshwater wetlands. What nutrient cycle is different between the two ecosystems that causes this difference in smell?

Answer: Salt marshes contain a lot more sulfur compounds than freshwater wetlands due to the high concentration of sulfate in saltwater. When a lot of organic carbon is produced, the decomposition of the organic matter uses up all of the oxygen and results in the respiration (reduction) of sulfate which produces sulfides. The sulfides have a strong smell. Although some sulfides are present in freshwater, the concentration is very low compared to saltwater. In freshwater wetlands, anoxic decomposition produces more methane because of the lack of sulfate or other terminal electron acceptors.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 21.4

7) Why does the burning of fossil fuels cause an imbalance within the carbon cycle? What is the consequence of this imbalance? Include a diagram of the carbon cycle to illustrate your answer.

Answer: Answers should include a diagram similar to what is shown in Figure 20.1 (page 632). The natural carbon cycle remains in balance because the production of carbon dioxide by respiration is roughly balanced by the consumption of carbon dioxide by photosynthesis. Over geological time, a large amount of carbon has built up in the form of fossil fuels, soils, sediments, and rocks. This stored carbon is being rapidly released as carbon dioxide through the burning of fossil fuels. The burning of fossil fuels is generating carbon dioxide faster than photosynthesis can consume or store it, thus the concentration of carbon dioxide in the atmosphere is increasing. Over geological time, these processes will continue and a new homeostasis will be reached, but in the near term the increased carbon dioxide may cause significant global warming (climate change) that will impact human society.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 21.1

8) How are the Ca cycle and the C cycle linked in marine environments? How will increased carbon dioxide from anthropogenic activity affect the Ca and C cycling in marine environments?

Answer: Several eukaryotic phototrophic microorganisms form calcareous (calcium-containing) exoskeletons. These planktonic phototrophs are also very important primary producers and fix large amounts of carbon dioxide into organic carbon in the ocean. In addition, the exoskeletons are composed of CaCO_3 (calcium carbonate). Carbon dioxide gas is very soluble in the ocean. As the atmospheric concentration of carbon dioxide increases, more carbon dioxide dissolves in the ocean in the form of carbon acid, lowering the pH and dissolving calcium carbonate. The acidification of the oceans is predicted to reduce the rate of formation of calcareous shells, which may reduce primary production and have effects on multiple nutrient cycles (C, Ca, N).

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 21.6

9) Why is sulfur cycling more complex to study than other nutrient cycles such as the nitrogen cycle? How might this influence researchers whose primary research interest is in understanding sulfur cycling?

Answer: Most other nutrient cycles are either exclusively or primarily processed through living organisms, but transformations in certain sulfur compounds also readily occur abiotically. Sulfur also has many different redox states. Researchers studying sulfur cycling need to consider abiotic redox reactions along with whichever organisms they are looking into for a particular step in the sulfur cycle. This requires adding additional abiotic experiments in addition to the ones containing living organisms of interest.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 21.4

10) Describe how syntrophy helps explain why certain enrichments yield consortia and never an individual species.

Answer: Some catabolic processes are favorable only when two different species cooperate in such a way that the first step is catalyzed by one species while the second species catabolizes the product formed from the first species. In this way, an enriched syntrophic consortium develops, and the process cannot be accomplished by only one species.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.2